Online Appendix

Next Station? The Impact of Subway Network Integration on Travel Behavior: Evidence from São Paulo's Line 5 - Lilac

This document contains the supplementary material, including detailed spatial distribution maps and additional tables for robustness and heterogeneity checks.

Appendix A: Figures

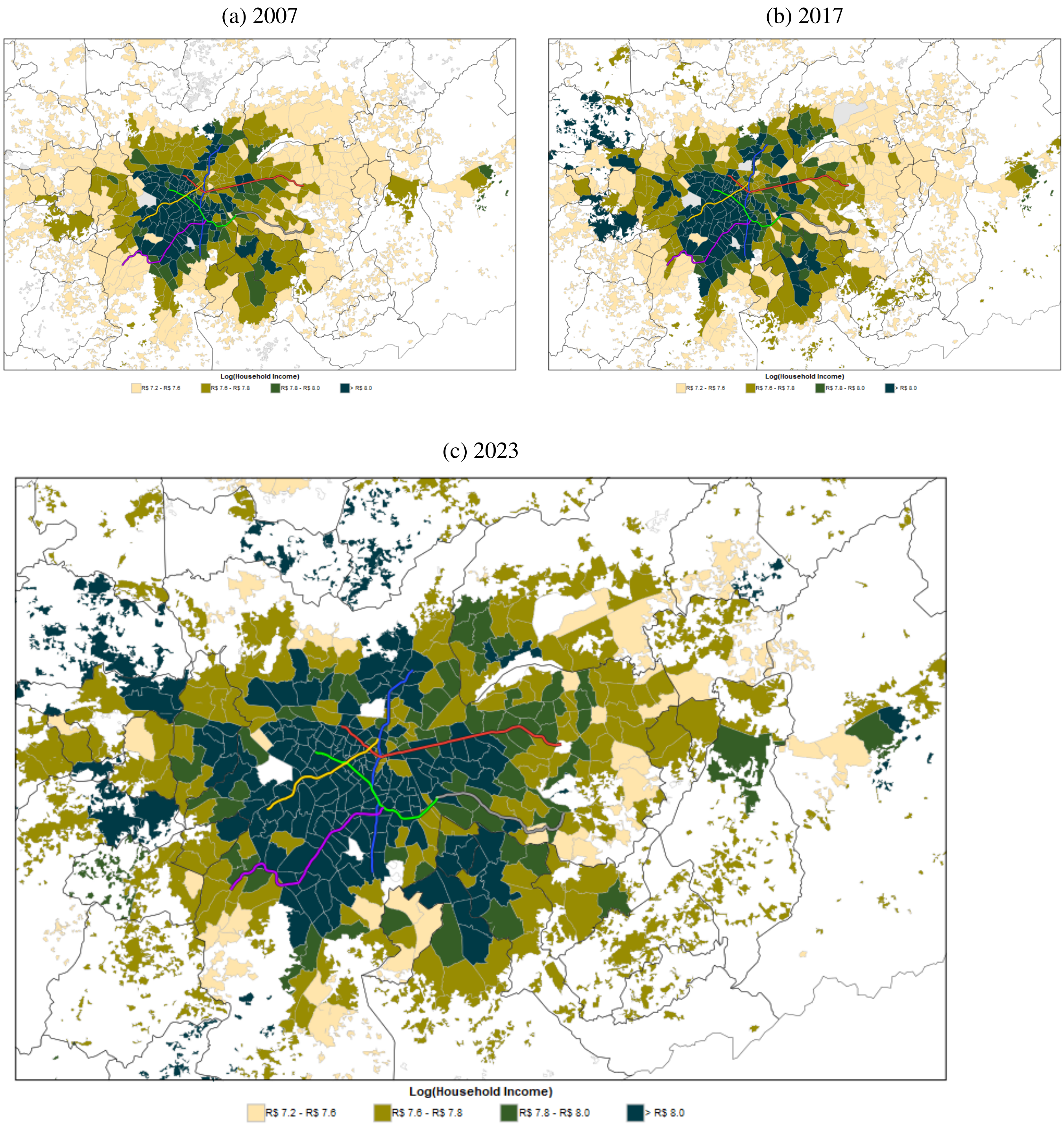


Figure 6: Spatial Distribution of Mean Income in the São Paulo Metropolitan Region

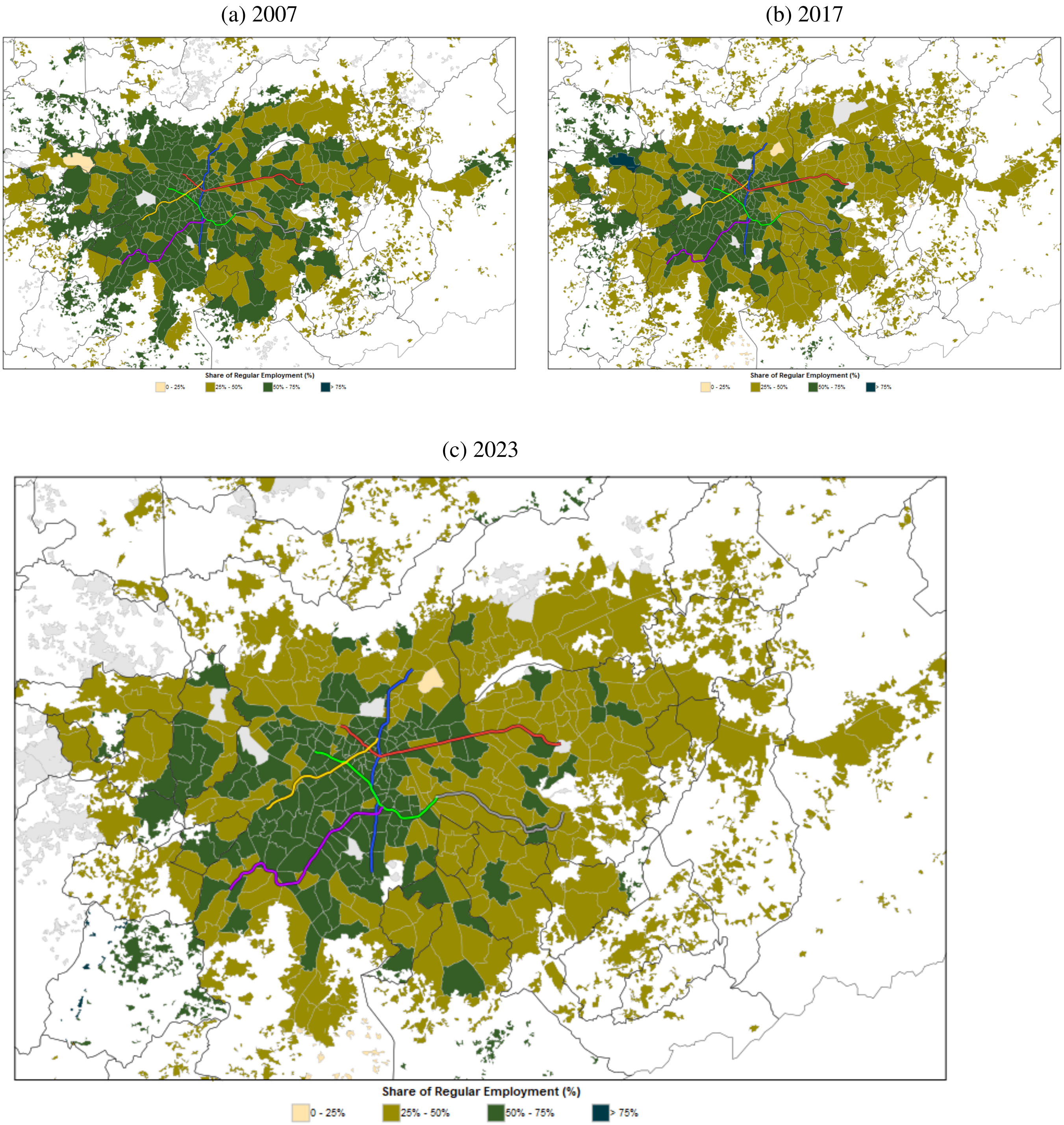


Figure 7: Spatial Distribution of Regular Employment in the São Paulo Metropolitan Region

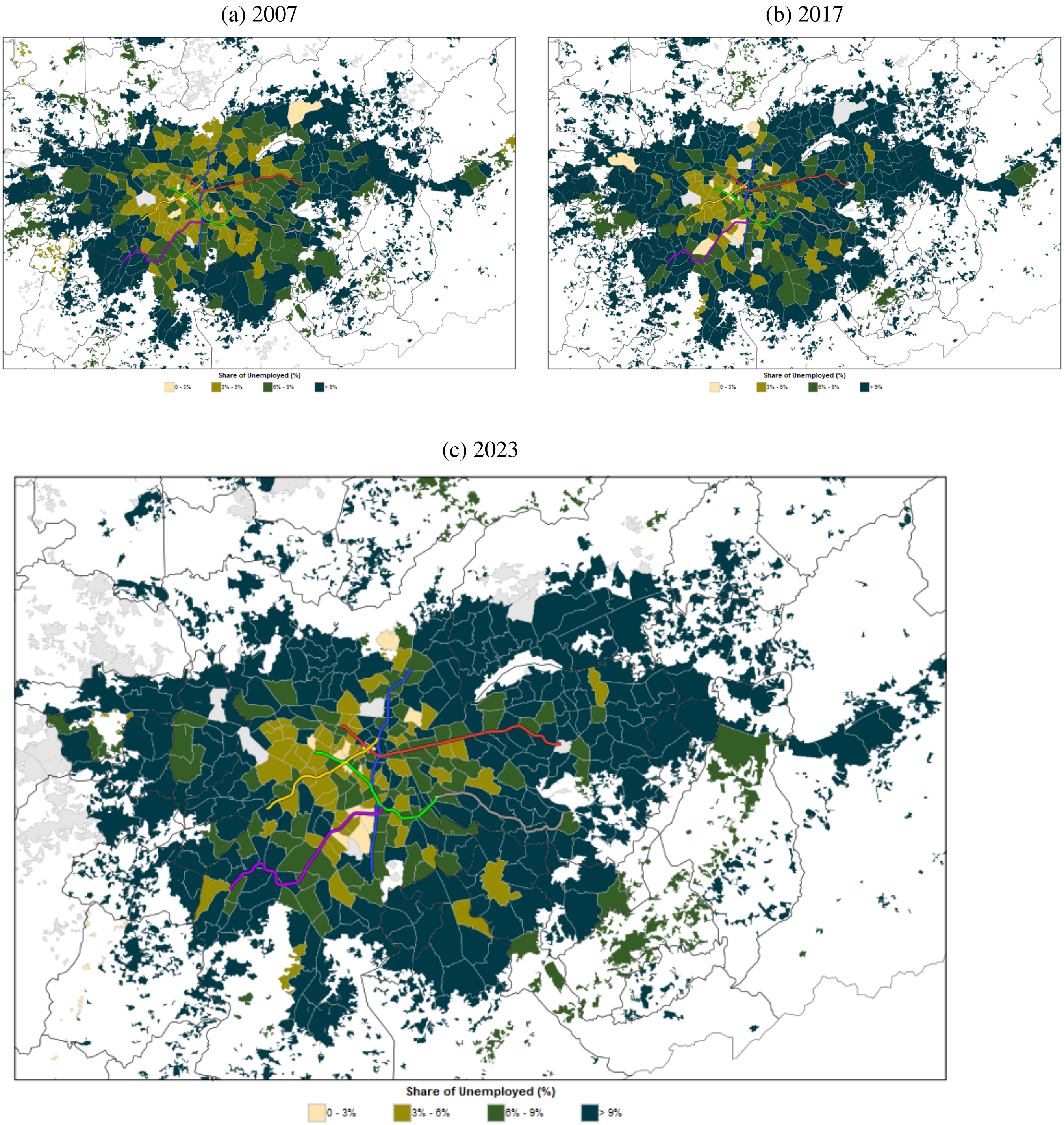


Figure 8: Spatial Distribution of Unemployement in the São Paulo Metropolitan Region

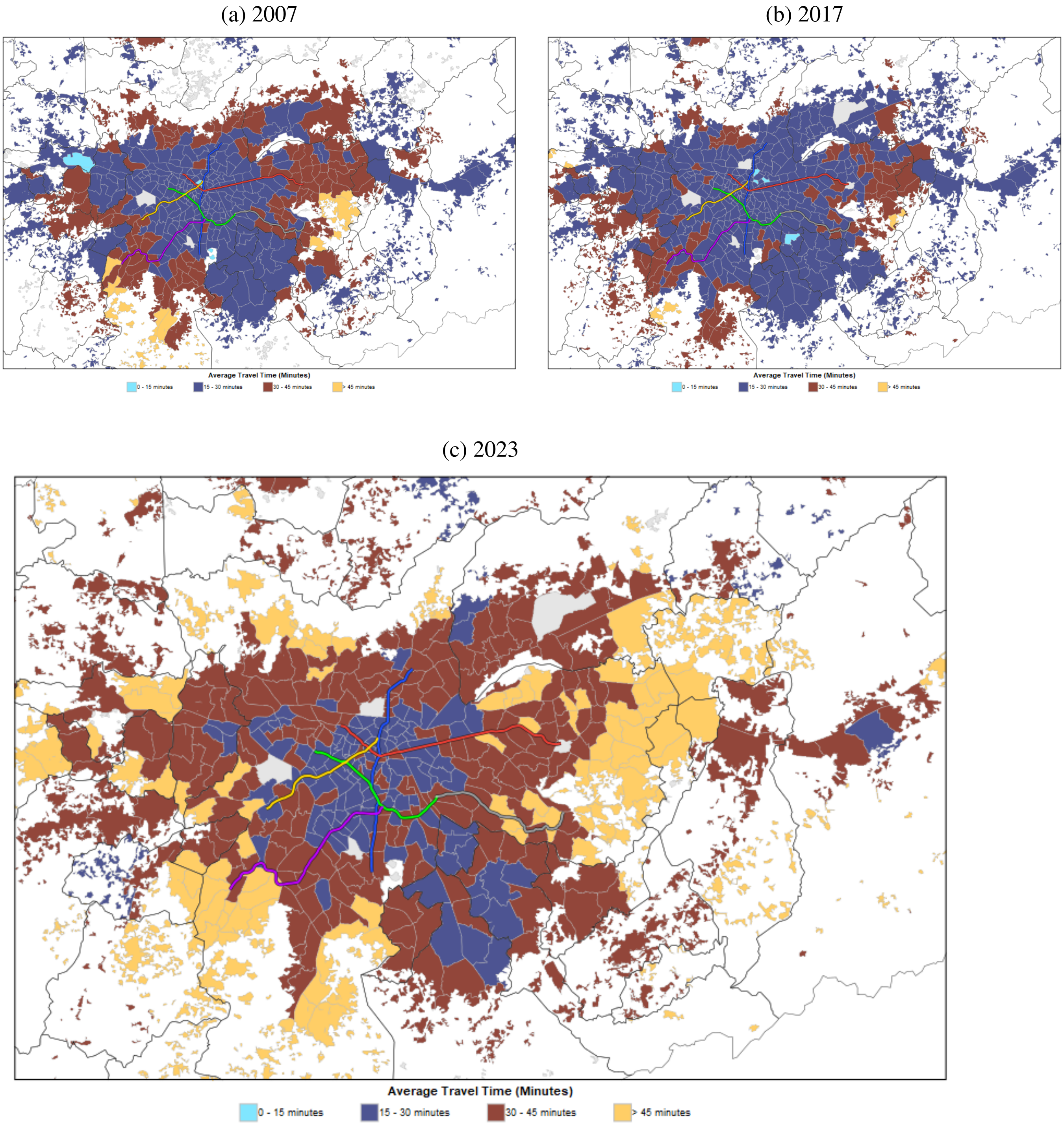
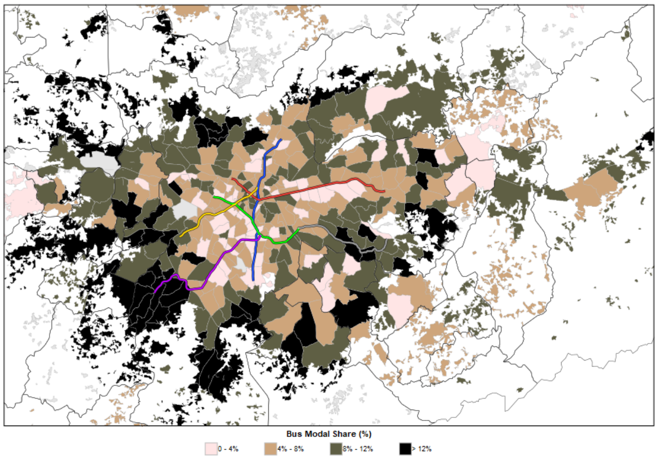
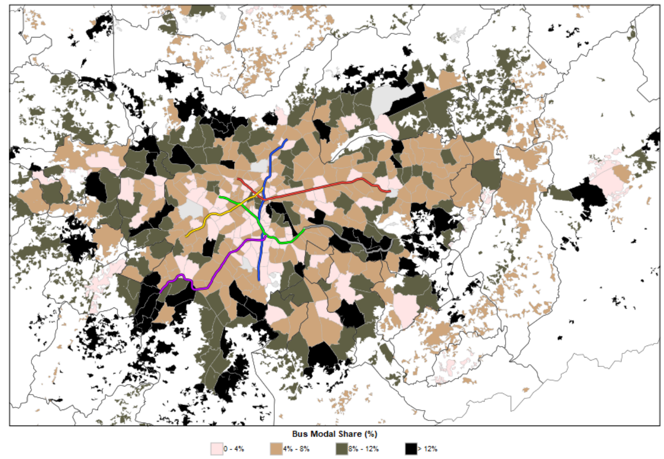


Figure 9: Spatial Distribution of Mean Trip Duration in the São Paulo Metropolitan Region

(a) 2007



(b) 2017



(c) 2023

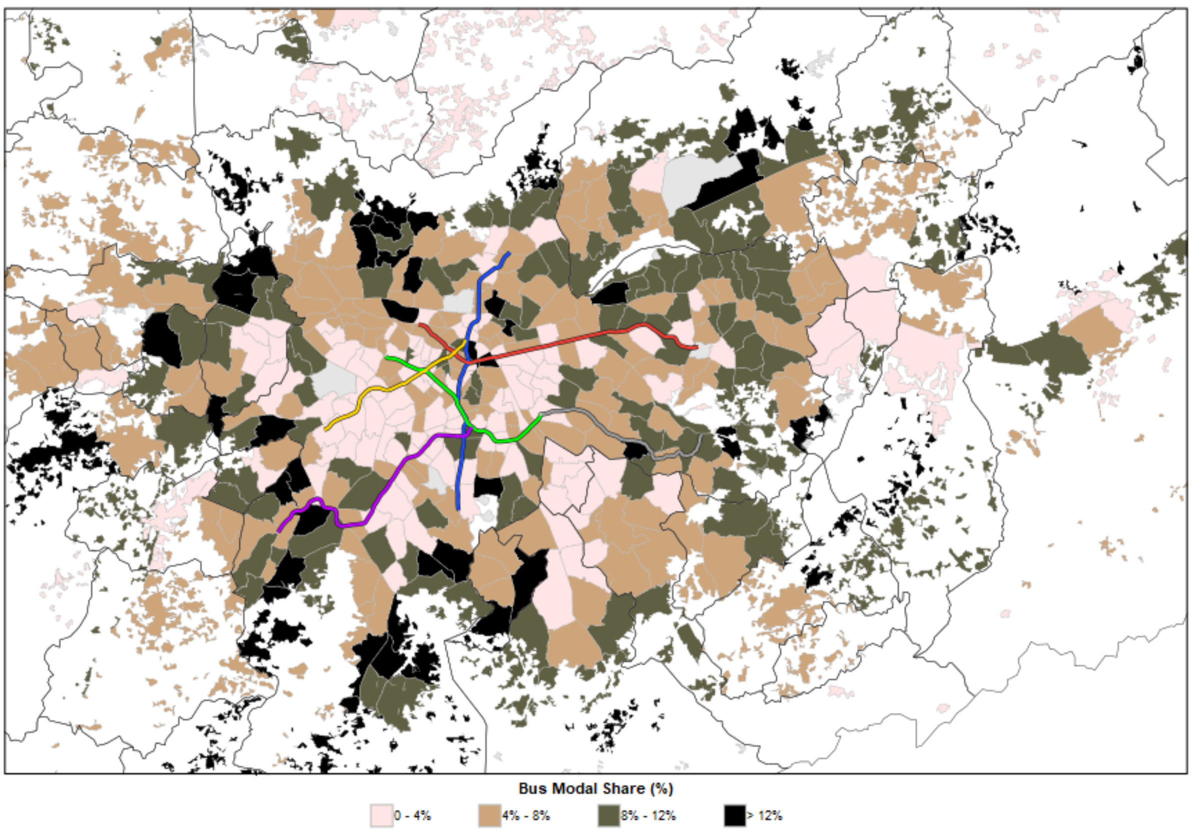


Figure 10: Spatial Distribution of Bus Modal Share in the São Paulo Metropolitan Region

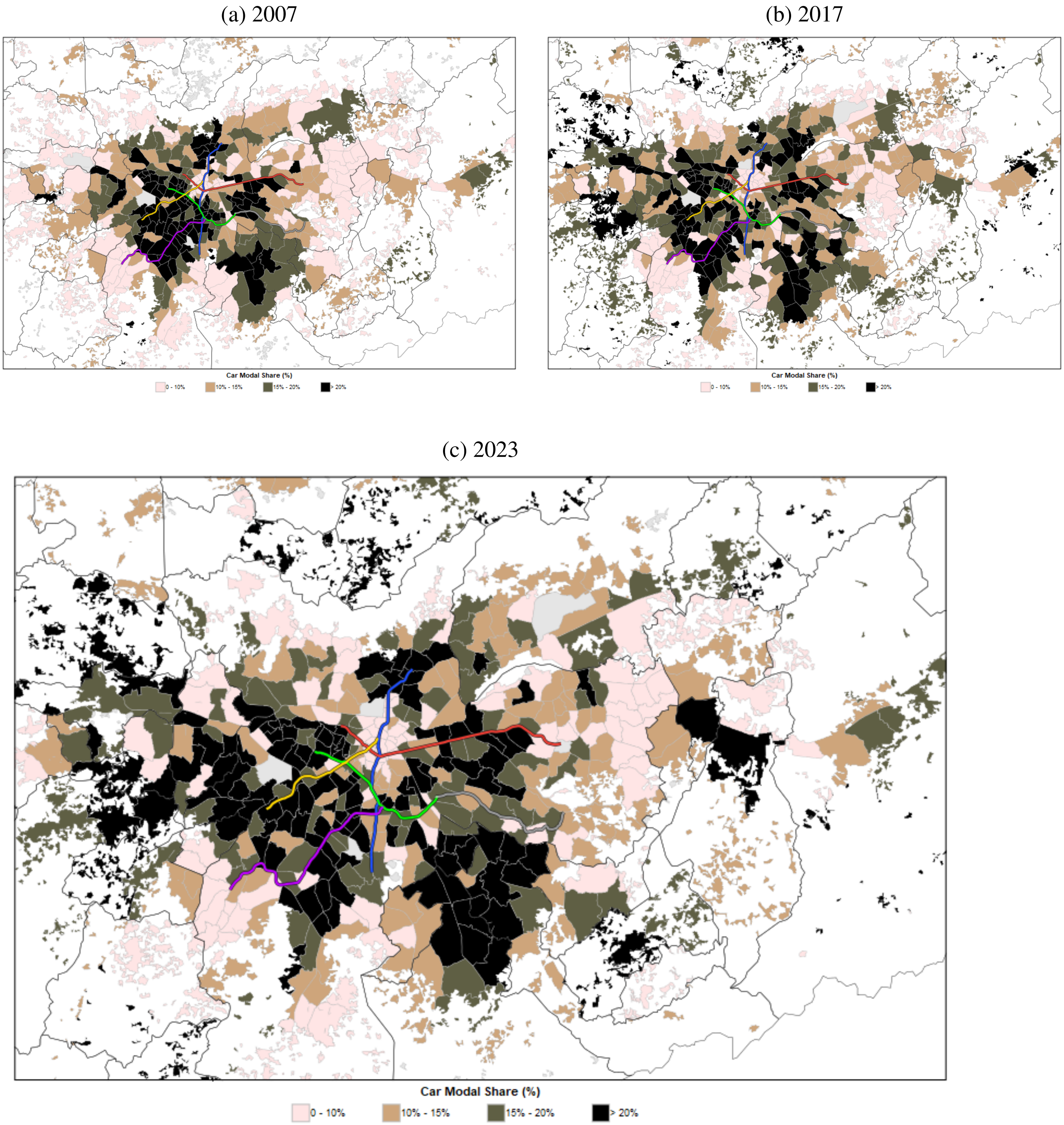


Figure 11: Spatial Distribution of Car Modal Share in the São Paulo Metropolitan Region

Appendix B: Tables

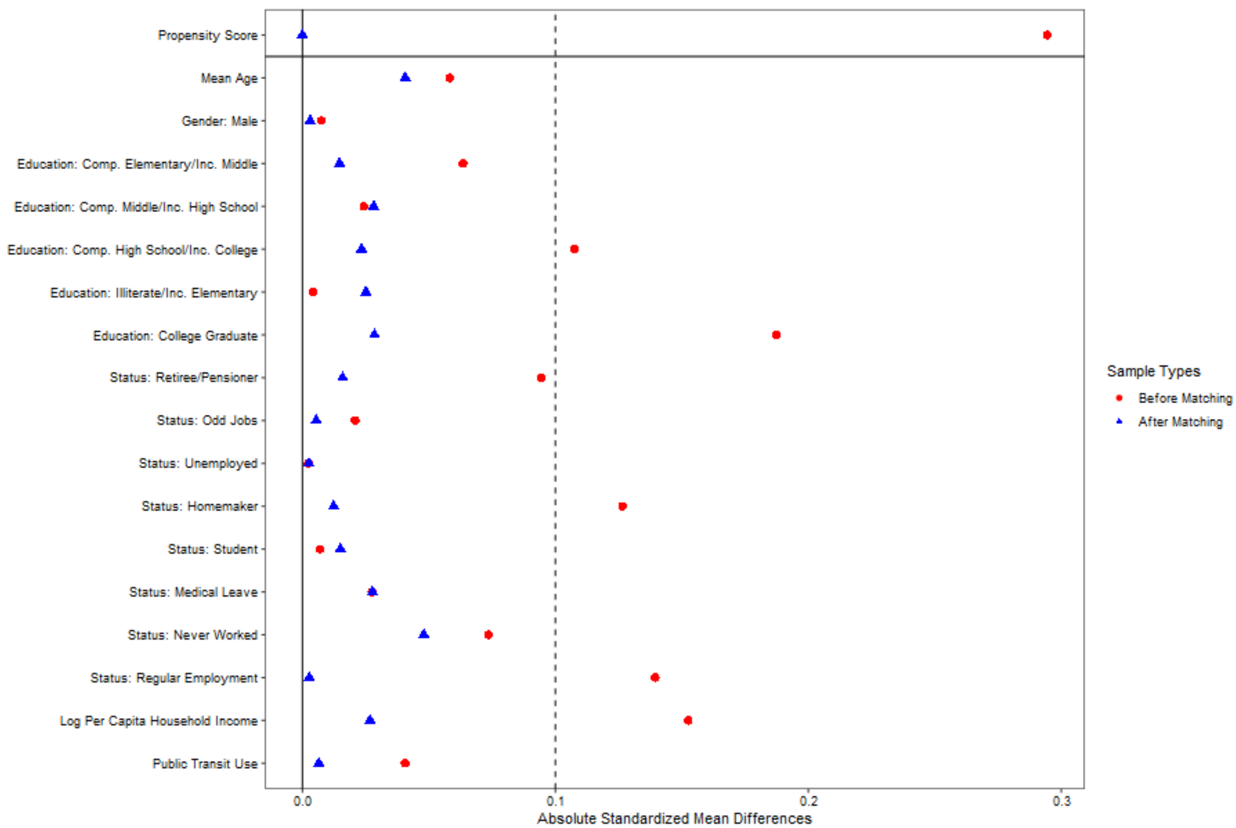


Figure 12: Love Plot of Covariate Balance Before and After Matching

Note: The plot illustrates the assessment of covariate balance using Absolute Standardized Mean Differences (ASMD). Red circles indicate the imbalance in the original, unmatched sample (Before Matching), whereas blue triangles demonstrate the bias reduction within the matched sample (After Matching). The vertical dashed line at 0.10 represents the conventional threshold for acceptable balance. Notably, all covariates were successfully adjusted, falling strictly below this benchmark.

Table 7: Event Study Results for Destination, Purpose, and Transport Mode - CPTM Control Group

	Control Group: CPTM									
	General		**Purpose**					**Mode**		
	Center	**Travels**	**Work**	**Study**	**Job**	**WkStJb**	**LzSpHe**	**Subway**	**Bus**	**Car**
Treatment × Post (2023)	0.011	-0.055**	0.016	0.008	-0.000	0.023	-0.016	0.044***	-0.009	-0.007
	(0.028)	(0.022)	(0.018)	(0.007)	(0.001)	(0.023)	(0.010)	(0.011)	(0.008)	(0.025)
Treatment × Pre (2007)	-0.008	0.021	0.007	0.002	-0.002	0.005	-0.011*	0.000	0.016*	-0.008
	(0.011)	(0.027)	(0.010)	(0.004)	(0.001)	(0.010)	(0.006)	(0.009)	(0.009)	(0.009)
Pre-treatment Mean	0.196	0.718	0.149	0.011	0.003	0.163	0.028	0.029	0.076	0.067
Relative Effect (%)	5.4	-7.7	11.0	70.3	-7.7	14.2	-55.6	153.3	-11.1	-10.1
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects										
MCZ	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,250	18,250	18,250	18,250	18,250	18,250	18,250	18,250	18,250	18,250

Note: The table consolidates the Event Study estimations comparing the Treatment group with the CPTM commuter rail control group. The columns represent different dependent variables: probability of traveling to the city center (Center), overall probability of making a trip (Travels), specific trip motivations (Work, Study, Job search, and the aggregates WkStJb and LzSpHe), and main transport modes (Subway, Bus, Car). WkStJb is the aggregate for Work, Study, or Job Search, while LzSpHe aggregates Leisure, Shopping, or Health. The "Treatment × Post (2023)" coefficient reports the post-intervention effect, while "Treatment × Pre (2007)" tests the parallel trends assumption. "Pre-treatment Mean" indicates the average proportion of each variable for the treatment group in the pre-intervention period (2007 and 2017). The "Relative Effect" is the ratio between the post coefficient and the pre-treatment mean. All specifications include individual covariates and fixed effects for MCZ and Year. Robust standard errors clustered at the MCZ level are in parentheses. Significance levels: 1% (***), 5% (**), and 10% (*).

Table 8: Event Study Results for Destination, Purpose, and Transport Mode - PSM-Matched Control Group

	Control Group: PSM-Matched									
	General		**Purpose**					**Mode**		
	Center	**Travels**	**Work**	**Study**	**Job**	**WkStJb**	**LzSpHe**	**Subway**	**Bus**	**Car**
Treatment × Post (2023)	0.034	0.004	0.034*	0.008	-0.002	0.041*	-0.007	0.036***	-0.008	-0.004
	(0.029)	(0.022)	(0.020)	(0.008)	(0.001)	(0.024)	(0.009)	(0.013)	(0.011)	(0.024)
Treatment × Pre (2007)	0.017	0.034	0.026**	0.001	-0.001	0.027*	-0.010	-0.004	0.023**	0.002
	(0.015)	(0.025)	(0.013)	(0.005)	(0.001)	(0.014)	(0.007)	(0.010)	(0.010)	(0.012)
Pre-treatment Mean	0.196	0.718	0.149	0.011	0.003	0.163	0.028	0.029	0.076	0.067
Relative Effect (%)	17.2	0.6	22.7	73.0	-76.9	25.0	-23.7	123.7	-10.9	-6.4
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects										
MCZ	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,496	10,496	10,496	10,496	10,496	10,496	10,496	10,496	10,496	10,496

Note: The table consolidates the Event Study estimations comparing the Treatment group with the PSM-Matched control group. The columns represent different dependent variables: probability of traveling to the city center (Center), overall probability of making a trip (Travels), specific trip motivations (Work, Study, Job search, and the aggregates WkStJb and LzSpHe), and main transport modes (Subway, Bus, Car). WkStJb is the aggregate for Work, Study, or Job Search, while LzSpHe aggregates Leisure, Shopping, or Health. The "Treatment × Post (2023)" coefficient reports the post-intervention effect, while "Treatment × Pre (2007)" tests the parallel trends assumption. "Pre-treatment Mean" indicates the average proportion of each variable for the treatment group in the pre-intervention period (2007 and 2017). The "Relative Effect" is the ratio between the post coefficient and the pre-treatment mean. All specifications include individual covariates and fixed effects for MCZ and Year. Robust standard errors clustered at the MCZ level are in parentheses. Significance levels: 1% (***), 5% (**), and 10% (*).

Table 9: Heterogeneity Analysis - Event Study on Subway Modal Share (CPTM Control Group)

Y = Subway	**Control Group: CPTM**								
	Sample Heterogeneity								
	Women (1)	**Men (2)**	**Low Income (3)**	**Youth (4)**	**Working Age (5)**	**Students (6)**	**Employed (7)**	**Zero-Vehicle (8)**	**Vehicle Owners (9)**
Treatment × Post (2023)	0.041^{***}	0.048^{***}	0.059^{**}	0.042^{**}	0.056^{***}	0.024	0.066^{***}	0.069^{***}	0.031^{**}
	(0.011)	(0.014)	(0.023)	(0.016)	(0.012)	(0.023)	(0.018)	(0.017)	(0.013)
Treatment × Pre (2007)	0.006	-0.005	-0.010	0.002	0.008	0.025	-0.013	-0.009	0.007
	(0.011)	(0.011)	(0.009)	(0.016)	(0.012)	(0.018)	(0.017)	(0.015)	(0.009)
Pre-treatment Mean	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029
Relative Effect (%)	140.7	163.8	202.8	143.4	194.5	82.8	227.9	238.3	108.3
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects									
MCZ	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,630	8,389	9,033	3,480	10,904	1,377	7,599	6,760	11,260

Note: This table presents the heterogeneity analysis for the Event Study estimations regarding subway usage as the main transport mode, using the CPTM commuter rail group as control. The columns (1) through (9) represent different subsamples of the population based on sociodemographic and mobility characteristics. The "Treatment × Post (2023)" coefficient reports the treatment effect in the post-intervention period, while "Treatment × Pre (2007)" tests the parallel trends assumption. "Pre-treatment Mean" indicates the average proportion of subway use by individuals in the treatment group in the pre-intervention years (2007 and 2017). The "Relative Effect" is calculated by the ratio between the Post period coefficient and the pre-treatment mean. All specifications include individual covariates and fixed effects for MCZ (Minimum Comparable Zones) and Year. Robust standard errors clustered at the MCZ level are in parentheses. Asterisks denote statistical significance at the 1% (***), 5% (**), and 10% (*) levels.

Table 10: Heterogeneity Analysis - Event Study on Subway Modal Share (PSM-Matched Control Group)

Y = Subway	**Control Group: PSM-Matched**								
	Sample Heterogeneity								
	Women (1)	**Men (2)**	**Low Income (3)**	**Youth (4)**	**Working Age (5)**	**Students (6)**	**Employed (7)**	**Zero-Vehicle (8)**	**Vehicle Owners (9)**
Treatment × Post (2023)	0.019	0.033^{*}	0.045^{*}	0.015	0.030^{**}	0.018	0.048^{**}	0.046^{*}	0.020
	(0.015)	(0.017)	(0.024)	(0.021)	(0.015)	(0.036)	(0.020)	(0.024)	(0.014)
Treatment × Pre (2007)	-0.027^{**}	-0.019	-0.030^{***}	-0.031	-0.013	-0.043	-0.038^{*}	-0.031^{*}	-0.013
	(0.014)	(0.014)	(0.011)	(0.020)	(0.014)	(0.031)	(0.020)	(0.016)	(0.012)
Pre-treatment Mean	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029
Relative Effect (%)	65.9	112.1	154.5	51.4	103.4	61.4	166.6	159.0	69.3
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects									
MCZ	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,458	4,884	5,176	2,069	6,516	789	4,208	3,681	6,649

Note: This table presents the heterogeneity analysis for the Event Study estimations regarding subway usage as the main transport mode, using the PSM-Matched group as control. The columns (1) through (9) represent different subsamples of the population based on sociodemographic and mobility characteristics. The "Treatment × Post (2023)" coefficient reports the treatment effect in the post-intervention period, while "Treatment × Pre (2007)" tests the parallel trends assumption. "Pre-treatment Mean" indicates the average proportion of subway use by individuals in the treatment group in the pre-intervention years (2007 and 2017). The "Relative Effect" is calculated by the ratio between the Post period coefficient and the pre-treatment mean. All specifications include individual covariates and fixed effects for MCZ (Minimum Comparable Zones) and Year. Robust standard errors clustered at the MCZ level are in parentheses. Asterisks denote statistical significance at the 1% (***), 5% (**), and 10% (*) levels.